

Hybrid Quantum-Classical Framework for Multi-Label Bengali Emotion Classification

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Abstract. Multi-label emotion classification of under-resourced languages such as Bengali remains a challenging task because of limited labelled corpora, severe class imbalance, and the need to recognise multiple co-occurring emotions per comment. This paper investigates hybrid quantum-classical frameworks that combine pretrained transformer representations with trainable variational quantum circuits (VQCs) for multi-label emotion classification on the EmoNoBa benchmark dataset. Along with experimenting with template-based VQCs such as the strongly entangling layer, we design two custom VQCs: a cross-coupling entangling scheme and a re-uploading encoding scheme with a rotating ansatz, and evaluate them with two backbone transformer encoders: IndicBERTv2 and BanglaBERT across 2-, 4-, and 8-qubit setups. For rigorous evaluation, in addition to the standard macro precision, recall, and F1 scores, we report the Exact Match Ratio (EMR) and an instance-level partial-match precision, recall, and F1-score. The quantum variants substantially improve the weaker BanglaBERT baseline, raising its macro F1-score from 0.2101 to 0.4850 and its EMR from 0.0000 to 0.4630, while the best overall macro F1-score of 0.5165 and the highest EMR of 0.4577 are achieved with IndicBERTv2 and Custom 2 at eight qubits. Our results indicate that the benefit of additional qubits depends critically on the compatibility of the encoding, entanglement pattern, and classical representation.

Keywords: hybrid quantum-classical machine learning · variational quantum circuits · multi-label emotion classification · Bengali natural language processing · low-resource languages

1 Introduction

Multi-label emotion classification is an important natural language processing (NLP) task with applications in sentiment-aware dialogue systems, social-media

analysis, public-opinion monitoring, and human-computer interaction. Unlike binary sentiment analysis, which assigns a single polarity, multi-label emotion recognition requires a model to predict a set of emotions for each input, since closely related emotions frequently co-occur. The problem becomes substantially more difficult for low-resource languages such as Bengali, because labelled corpora, language-specific pretrained models, balanced emotion datasets, and standardized benchmarks are comparatively limited [3,8]. Minority emotion classes are easily suppressed by dominant labels, and the informal, noisy style of social-media text compounds the difficulty.

In parallel, near-term quantum machine learning (QML) has emerged as a promising paradigm in which parameterized quantum circuits (PQCs) map classical feature vectors into high-dimensional quantum Hilbert spaces [7]. Variational quantum circuits (VQCs) with trainable rotation parameters updated by gradient descent have been applied to supervised learning [14], and the data re-uploading paradigm interleaves classical feature encoding layers $U(x)$ with trainable unitaries $U(\theta)$ to increase the expressivity of low-qubit circuits [13]. Recent analyses of parameterized circuit expressibility and entanglement [16] and of hybrid quantum-classical training dynamics [1] provide a theoretical basis for designing trainable quantum feature extractors. These developments motivate the question of whether a VQC, inserted as a trainable bottleneck inside a classical transformer architecture, can improve multi-label emotion classification for Bengali.

In this paper, we present a hybrid quantum-classical framework that combines a pretrained transformer encoder with a trainable VQC for multi-label emotion classification. The contributions of this paper are as follows:

- We develop a hybrid quantum-classical framework that couples a pretrained transformer encoder (IndicBERTv2 [5] or BanglaBERT [3]) with a trainable VQC for multi-label emotion classification, and a classical baseline without a quantum branch.
- We design and compare three VQC architectures: 1) strongly entangling layer (SEL), 2) cross-coupling entangling scheme (VQC 1) and 3) re-uploading encoding scheme with a rotating ansatz (VQC 2), and evaluate both across 2, 4-, and 8-qubit setups.
- We evaluate all configurations with a unified protocol covering macro, precision, recall, and F1-scores, the Exact Match Ratio (EMR), and an instance-level partial-match precision, recall, and F1-score, and analyse how encoding, entanglement, and circuit depth interact with the quality of the classical representation.

2 Related Work

Bengali remains comparatively underrepresented in NLP despite its large speaker population, with limited annotated resources and language-specific models historically constraining downstream tasks such as emotion and sentiment analysis [4]. The development of pretrained models such as BanglaBERT and In-

dicBERTv2 has substantially improved Bengali and Indic language understanding [3,5]. For emotion analysis, EmoNoBa [8] introduced a manually annotated dataset of noisy Bangla social-media comments with six fine-grained, non-exclusive emotion categories, making it suitable for multi-label emotion classification. Multi-label learning differs from multi-class classification by allowing multiple labels per instance, and its established formulations and challenges have been comprehensively studied by Tsoumakas and Katakis [17] and Zhang and Zhou [20].

Quantum approaches have recently attracted attention in NLP, including sentiment analysis, text classification, sequence labelling, and multimodal affective computing [19,6,18]. Quantum probability and quantum-inspired models have been applied to sentiment and emotion-related tasks [?,?,9], while variational quantum circuits (VQCs) have enabled hybrid classical–quantum approaches for multimodal sentiment analysis [?]. Quantum recurrent architectures have also been investigated for NLP tasks such as POS tagging and text classification [?,?,?]. For Bengali specifically, Pal et al. [12] explored a hybrid classical–quantum framework for sentiment classification and claim check-worthiness identification, highlighting the emerging potential of quantum-enhanced methods for Bengali NLP.

VQCs constitute a central component of near-term quantum machine learning, combining parameterized quantum operations with classical optimization [2,11,14]. Their performance depends strongly on circuit architecture, entanglement structure, and data-encoding strategy [16,15]. At the same time, barren-plateau phenomena can hinder the training of deep or poorly designed circuits, motivating shallow and structured ansatz designs [10]. Despite these developments, the application of hybrid transformer–VQC architectures to multi-label emotion detection in noisy Bengali text remains largely unexplored. Motivated by this gap, we investigate a quantum-enhanced framework that combines pre-trained Bengali contextual representations with trainable VQCs for multi-label emotion classification on EmoNoBa, with particular attention to class imbalance and label-wise decision thresholds.

3 Dataset

This study uses the EmoNoBa dataset, a manually annotated collection of 22,698 public Bangla comments gathered from social media platforms [8]. The comments span 12 domains, including personal matters, politics, and health, and are labelled with six fine-grained emotion categories based on the Junto Emotion Wheel. The dataset preserves the informal and noisy characteristics of online Bangla text, making it suitable for evaluating emotion-classification models under realistic conditions.

The dataset is split into training, validation, and test partitions of 18,420, 2,047, and 2,272 comments, respectively. Table 1 reports the per-emotion label counts in the training partition. The distribution is strongly imbalanced: *Joy*

Table 1: Distribution of emotion labels in the training partition of the EmoNoBa dataset.

Emotion Count Percentage		
Joy	8,314	45.14
Sadness	4,569	24.80
Love	3,786	20.55
Anger	3,542	19.23
Surprise	848	4.60
Fear	279	1.51
Total	18,420	100.00

(8,314) dominates, whereas *Surprise* (848) and especially *Fear* (279) are severely underrepresented.

4 Methodology

4.1 Task Definition

Given a Bengali comment x consisting of a sequence of tokens, the objective is to predict a binary label vector $y \in \{0, 1\}^K$ over $K = 6$ emotion categories: *Love*, *Joy*, *Surprise*, *Anger*, *Sadness*, *Fear*, where $y_k = 1$ indicates that emotion k is expressed in the comment and multiple emotions may co-occur. Following the convention of multi-label classification, the model produces an independent logit z_k for each emotion, which is converted to a probability via the sigmoid function $\sigma(z_k) = 1/(1 + e^{-z_k})$ and thresholded at 0.5 to obtain the predicted label $\hat{y}_k$. The goal is to maximise joint subset-prediction accuracy, measured by the Exact Match Ratio, while retaining competitive per-label performance.

4.2 Framework Description

Figure 1 provides an overview of the proposed hybrid classical-quantum framework. For both the classical baseline and the quantum variants, each comment is first tokenized with a maximum sequence length of 128 and passed through a pretrained transformer encoder, either IndicBERTv2 or BanglaBERT, from which the 768-dimensional [CLS] representation is extracted. A trainable linear layer projects this representation to a 128-dimensional feature vector, which is then squashed by a tanh activation.

In the quantum variant, the compressed features are further projected by a linear layer into a low-dimensional vector matching the qubit register (e.g. four features for a four-qubit circuit) and fed into the VQC described in Section 4.3. The expectation values produced by the circuit are mapped back through a post-quantum linear layer to the 128-dimensional working space, followed by a ReLU

activation. In the classical baseline, the quantum block is removed, and the 128-dimensional features are passed directly to the classifier. In both cases, a final linear layer maps the representation to the $K = 6$ logits, which are interpreted with sigmoid outputs for multi-label prediction.

4.3 VQC Designs

Following the transformer encoding, the 768-dimensional [CLS] token embedding is projected by a trainable linear layer to a low-dimensional feature vector. Depending on the variant, the vector is squashed by $\tanh$ and fed directly into a four-qubit variational quantum circuit (VQC); the circuit outputs are mapped back through a post-quantum linear layer and a ReLU activation before the final classifier. Three VQCs are investigated: a strongly-entangling circuit (SEL), and two custom designs, VQC 1 and VQC 2.

SEL reference VQC: This circuit acts on $n_q \in \{2, 4, 8\}$ qubits and consumes an $\mathbb{R}^{n_q}$ vector. The first phase in this VQC is data encoding, where each feature is directly injected as a single-axis rotation $R_X(x_i)$ on qubit i (angle embedding). Next is the variational ansatz, where a single variational layer of PennyLane’s **StronglyEntanglingLayers** is applied, which places a general single-qubit rotation on every qubit and then entangles all qubits with CNOT gates, producing $3n_q$ trainable parameters. The final block is the measurement block, where the Pauli-Z expectation is measured on every qubit, producing n_q outputs for the post-quantum linear layer. The diagrammatic representation of SEL VQC is provided in Figure 2.

Cross-coupling VQC (VQC 1): The diagrammatic representation of 4-qubit VQC 1 is provided in Figure 3. In this circuit, each classical feature is directly injected as a single-axis rotation $R_Y(x_i)$ on qubit i (data encoding via angle embedding). Following data encoding, two variational layers are used, each consisting of a general rotation applied to every qubit followed by a controlled-NOT entangling stage. The first stage uses a nearest-neighbour ring, with CNOT gates on $(0, 1)$, $(1, 2)$, $(2, 3)$, and $(3, 0)$; the second uses a cross-coupling pattern, with CNOT gates on $(0, 2)$, $(1, 3)$, $(2, 1)$, and $(3, 0)$. The same 12-parameter weight tensor is shared by both Rot layers. Next is the measurement block, which is the same as SEL VQC.

Re-uploading VQC (VQC 2): The diagrammatic representation of VQC 2 is provided in Figure 4. The components of this VQC are as follows: 1) Data Encoding: For each qubit q , its feature pair (x_{2q}, x_{2q+1}) is first rotated by a learnable phase ϕ_q :

$$x'_{2q} = s_q(\cos \phi_q x_{2q} - \sin \phi_q x_{2q+1}), \quad (1)$$

$$x'_{2q+1} = s_q(\sin \phi_q x_{2q} + \cos \phi_q x_{2q+1}) \quad (2)$$

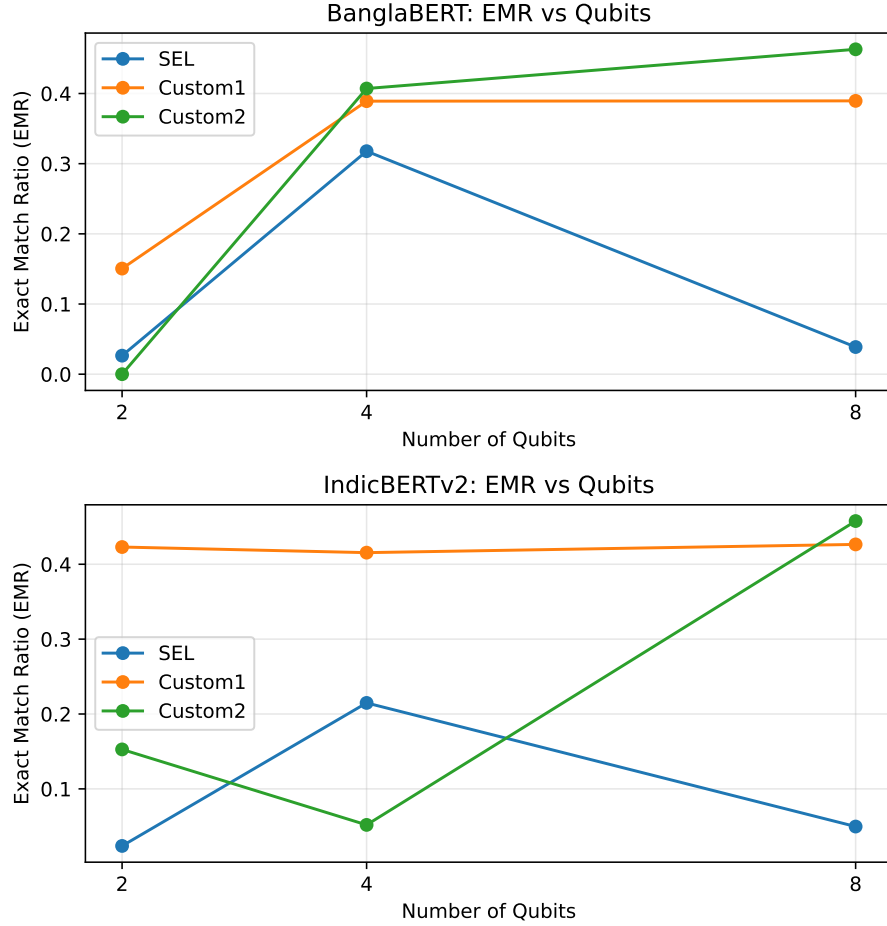


Fig. 1: End-to-end framework of the hybrid classical-quantum model for multi-label Bengali emotion classification. The dashed classical path corresponds to traditional pipelines, where the compressed text representation is fed directly to the classifier without any quantum component; it is shown for reference and is not a branch in our model. The solid quantum path inserts the variational quantum circuit between the compressed text representation and the post-quantum fusion layer, replacing the direct classical connection in the proposed framework.

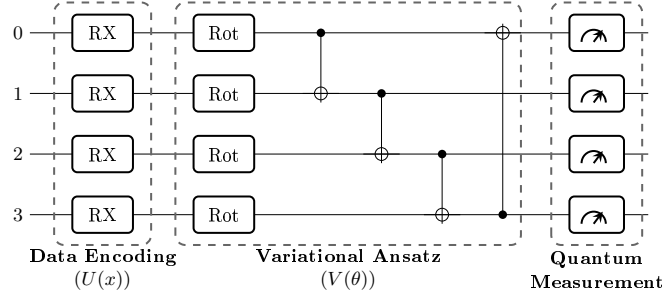


Fig. 2: Four-qubit circuit architecture used for the SEL VQC.

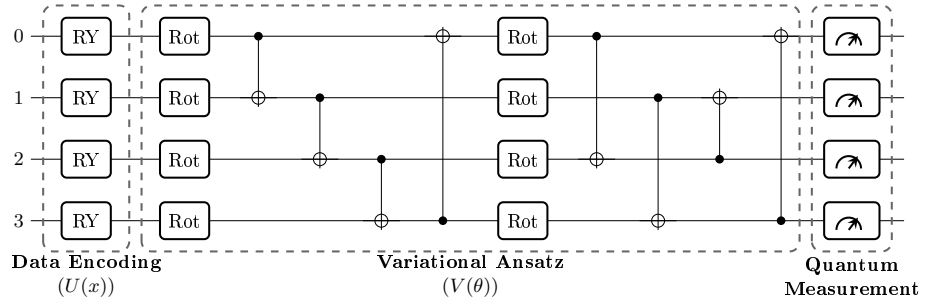


Fig. 3: Four-qubit circuit architecture used for the Custom 1 VQC.

where s_q is a second learnable amplitude per qubit. The rotated values are applied as rotation angles $R_X(x'_{2q})$ and $R_Y(x'_{2q+1})$ on qubit q . 2) Ansatz: Each variational block applies a general single-qubit rotation $R_Z R_Y R_Z \equiv \text{Rot}$ on every qubit, followed by a ring of controlled- Z entangling gates $CZ[q, (q+1) \bmod 4]$. 3) Data re-uploading: The data-dependent encoding is inserted before both variational blocks, i.e. $\text{encode} \rightarrow \text{ansatz} \rightarrow \text{encode} \rightarrow \text{ansatz}$, so the input is re-supplied between trainable layers. 4) Measurement: Both $\langle Z_q \rangle$ and $\langle X_q \rangle$ are measured on all four qubits, yielding $2n = 8$ expectation values that feed the post-quantum layer.

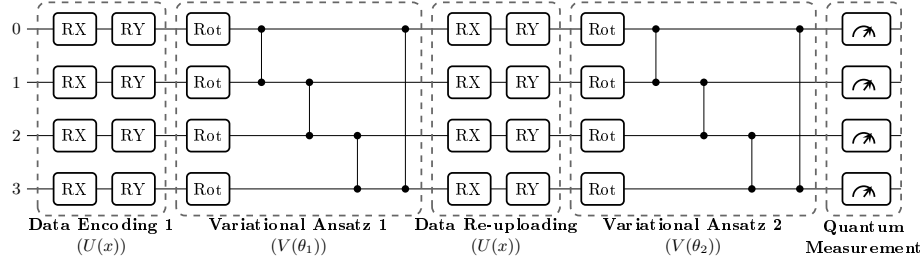


Fig. 4: Four-qubit circuit architecture used for the Custom 2 VQC.

4.4 Computational and Circuit Complexity

Let n_q denote the number of qubits, L the number of variational blocks (with the associated data re-upload), and B the mini-batch size. The variational quantum circuit is simulated exactly on the PennyLane `default.qubit` statevector simulator, which represents the quantum state as a complex vector of dimension 2^{n_q} . Applying any single gate to the full statevector costs $\mathcal{O}(2^{n_q})$ operations.

For the Custom 2 circuit (one ansatz layer, data re-uploading performed twice), the gate count is:

- **Encoding block:** 2 rotations per qubit (R_X and R_Y) = $2n_q$ gates.
- **Ansatz block:** n_q single-qubit Rot + n_q CZ = $2n_q$ gates.
- **Total (2 repetitions):** $2(2n_q + 2n_q) = 8n_q$ gates.

Hence a single circuit evaluation has asymptotic time complexity

$$T_{\text{circuit}} = \mathcal{O}(n_q 2^{n_q}), \quad (3)$$

and, because the hybrid network executes the circuit once per sample in a mini-batch of size B , the batch execution cost scales as

$$T_{\text{batch}} = \mathcal{O}(B n_q 2^{n_q}). \quad (4)$$

In contrast, the SEL reference circuit uses n_q single-axis rotations (R_X), n_q general Rot gates, and n_q CNOT gates ($3n_q$ gates total), while Custom 1 uses

n_q R_Y gates followed by two variational stages of n_q rotations and n_q CNOT gates ($5n_q$ gates total), both without feature re-uploading.

The dominant computational cost of a full forward pass is nevertheless the transformer encoder. For a sequence length S , hidden dimension d , and L_t layers, the classical self-attention complexity is

$$T_{\text{BERT}} = \mathcal{O}(B L_t S^2 d), \quad (5)$$

which, for the large pretrained models evaluated here (e.g., IndicBERTv2 with $\approx 278\text{M}$ parameters), heavily outweighs the quantum statevector simulation in wall-clock time. In addition, the quantum statevector memory footprint remains negligible: $16 \cdot 2^{n_q}$ bytes (e.g., 4KB at $n_q = 8$).

4.5 Training

All models were trained for a maximum of 25 epochs with a batch size of 8 and a maximum sequence length of 128. The AdamW [?] optimizer was used with a learning rate of 3×10^{-5} and a weight decay of 0.01. Because the EmoNoBa emotion labels are strongly imbalanced (Table 1), the binary cross-entropy loss was weighted with per-label positive weights computed from the inverse label frequency: *Love* 3.87, *Joy* 1.22, *Surprise* 20.72, *Anger* 4.20, *Sadness* 3.03, and *Fear* 65.02. This up-weights the gradient contribution of the rare *Fear* and *Surprise* classes so that they are not dominated by the frequent labels.

The learning rate was annealed with a ReduceLROnPlateau scheduler (factor 0.5, patience 2, minimum 10^{-6}) monitoring the validation loss; gradients were clipped to a maximum norm of 1.0, and training was stopped early if the validation macro-F1 did not improve for five consecutive epochs, restoring the best model state. The quantum circuit parameters and the classical weights were optimized jointly in a single backward pass.

4.6 Evaluation Metrics

We evaluate all configurations with three complementary schemes. *Scheme 1* is the per-label evaluation based on the macro precision, recall, and F1-scores. *Scheme 2* is the *Exact Match Ratio* (EMR), which scores a sample as correct only if its entire predicted label vector matches the gold vector:

$$\text{EMR} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}_i = y_i]. \quad (6)$$

Scheme 3 is an instance-level *partial-match* score that gives partial credit when the predicted set of positive labels overlaps the gold set without being identical. For each sample, the active label sets $G_i = \{k : y_{ik} = 1\}$ and $P_i = \{k : \hat{y}_{ik} = 1\}$ are intersected, and per-instance precision, recall, and F1 are computed as

$$P_i = \frac{|G_i \cap P_i|}{|P_i|}, \quad R_i = \frac{|G_i \cap P_i|}{|G_i|}, \quad F1_i = \frac{2P_i R_i}{P_i + R_i}, \quad (7)$$

with the edge cases that both empty sets yield a perfect score of 1 and a one-empty set yields 0. The reported partial-match precision, recall, and F1 are the macro averages of P_i , R_i , and $F1_i$ over the test set.

Single-label partitioning. Test instances can be partitioned by label cardinality into single-label samples ($\sum_{k=1}^K y_{ik} = 1$), representing mutually exclusive emotions, and multi-label samples ($\sum_{k=1}^K y_{ik} > 1$). Filtering the EmoNoBa test set for single-label instances yields $N_{\text{single}} = 940$ samples (41.4% of the test set). On this partition, strict Exact Match Ratio (EMR), Micro-F1, precision, and recall converge to standard multi-class classification accuracy, providing a clean diagnostic subset to evaluate class-prediction collapse versus boundary separability.

Statistical significance testing. To determine whether differences in Exact Match Ratio are statistically significant, we conduct hypothesis tests across the test set of $N = 2,272$ samples (284 batches $\times$ 8 samples/batch). Each configuration produces an aggregate EMR $= \hat{p} = x/N$. We compare proportions using the two-proportion z -test with pooled variance:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_{\text{pool}}(1 - \hat{p}_{\text{pool}}) \left(\frac{1}{N} + \frac{1}{N} \right)}}, \quad \hat{p}_{\text{pool}} = \frac{x_1 + x_2}{2N}, \quad (8)$$

reporting the two-sided p -value $p = 2\Phi(-|z|)$ at an $\alpha = 0.05$ significance threshold. We note the methodological caveat that because all configurations share the same test set, predictions exhibit positive correlation; the two-proportion z -test assumes independent samples, thus underestimating inter-model correlation, but serves as a principled aggregate significance test on shared test proportions.

5 Experiments and Results

5.1 Experimental Setup

All experiments were performed using the PyTorch and PennyLane libraries. The transformer encoders ran on a pair of NVIDIA T4 GPUs, while the variational quantum circuits were executed on a CPU using PennyLane’s `default.qubit` simulator, with circuit outputs moved back to the GPU after each forward pass. A random seed of 42 was fixed across Python, NumPy, and PyTorch (including CUDA) for reproducibility. All models were evaluated on the same held-out test partition of 2,272 comments.

5.2 Noise-Aware Training Under the Depolarizing Channel

Models trained on ideal (noiseless) quantum simulators often suffer significant performance degradation when deployed on real quantum hardware due to gate-level noise. To evaluate the robustness of our best-performing 4-qubit IndicBERTv2 configuration, we retrain the model under simulated quantum noise using PennyLane’s `default.mixed` density-matrix simulator.

We employ the *depolarizing channel*, a standard single-qubit noise model that replaces a qubit state ρ with the maximally mixed state with probability p :

$$\mathcal{E}_p(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \quad (9)$$

where $p \in [0, 1]$ is the depolarizing probability and X, Y, Z are the Pauli matrices. At $p = 0$ the channel is noiseless; as p increases, the quantum information is progressively lost.

A depolarizing channel with $p = 0.05$ is inserted after *every* gate in the variational quantum circuit, as illustrated in Fig. 5. For single-qubit gates (R_Y , Rot), one channel is applied on the gate’s wire. For two-qubit CNOT gates, a channel is applied on both the control and target wires. This results in a total of 28 noise channels across the 20 gates in the circuit.

The simulation uses density matrices (16×16 for 4 qubits) instead of state vectors, which allows exact modelling of noise effects. Gradients are computed via backpropagation through the density-matrix evolution using PennyLane’s PyTorch interface. All other training settings remain identical to the ideal setup, allowing a direct comparison of model performance.

Robustness Results The results of the noise-aware training are presented in Table 2. When subjected to severe gate-level noise ($p = 0.05$), the 4-qubit IndicBERTv2 model demonstrates expected degradation in instance-level Macro and Partial Match F1-scores due to the loss of quantum coherence. However, interestingly, the Exact Match Ratio improves significantly under noisy conditions (from 0.4155 to 0.5528). This suggests that the depolarizing noise acts as a strong regularizer during training, forcing the model to make more decisive, synchronized predictions across all emotion labels, which is highly beneficial for strict multi-label classification tasks.

Table 2: Performance comparison of the 4-qubit IndicBERTv2 model under ideal (noiseless) and noisy ($p = 0.05$) simulated environments.

Framework	Qubit	Environment	Macro			Partial Match			Exact Match Ratio
			Precision	Recall	F1-score	Precision	Recall	F1-score	
IndicBERTv2	4	Ideal (VQC 1)	0.4616	0.5496	0.4922	0.5996	0.6845	0.6141	0.4155
		Noisy ($p = 0.05$)	0.4177	0.4409	0.4259	0.5528	0.5528	0.5528	0.5528

5.3 Results

The performance of the classical baseline and the quantum variants is reported in Table 3. For the IndicBERTv2 encoder, the classical baseline achieves a macro F1-score of 0.5333, a partial-match F1-score of 0.6268, and an EMR of 0.4252.

The SEL reference circuit does not surpass the classical baseline under the stronger IndicBERTv2 encoder. Its best macro F1-score of 0.4572 and its best

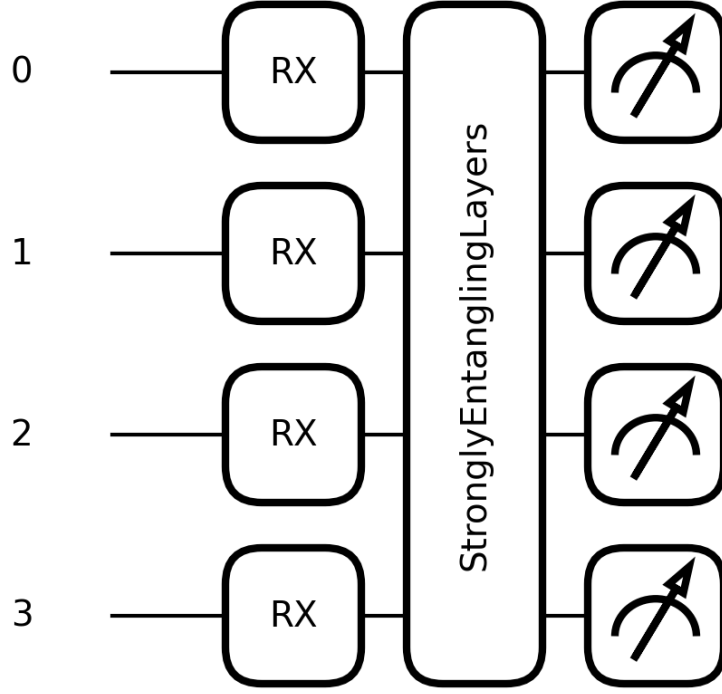


Fig. 5: Noise-aware variational quantum circuit with 4 qubits and depolarizing probability $p = 0.05$. Wires 0–3 represent the four qubits. R_Y gates encode compressed features from IndicBERTv2. Rot gates are trainable variational rotations. CNOT gates create entanglement in ring ($0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$) and cross ($0 \rightarrow 2, 1 \rightarrow 3, 2 \rightarrow 1, 3 \rightarrow 0$) topologies. Double vertical lines indicate structural barriers separating the encoding, variational, and entanglement stages. Each small box labelled $D(\mathbf{p})$ denotes a depolarizing channel $\mathcal{E}_{0.05}$ that simulates gate-level noise. The rightmost symbols denote Pauli- Z expectation value measurements.

EMR of 0.2148 are both obtained at four qubits, where the partial-match F1-score reaches 0.5387; at two and eight qubits, the EMR drops to 0.0238 and 0.0497, respectively. The single SEL thus captures only part of the benefit of the additional qubit capacity.

VQC 1 improves consistently as the qubit count grows from two to eight, raising the macro F1-score from 0.4869 to 0.4982 and the EMR from 0.4230 to 0.4265, with a partial-match F1-score of 0.6164 at two qubits and 0.6144 at eight qubits. Its EMR remains close to the classical 0.4252 at every qubit size, indicating that the cross-coupling entangling scheme preserves the quality of the strong classical representation without surpassing it.

Custom 2 achieves the strongest quantum results with IndicBERTv2 at eight qubits, where it reaches a macro F1-score of 0.5165, a partial-match F1-score of 0.6255, and the highest EMR of 0.4577, surpassing both the classical baseline and the other two circuits. At smaller qubit counts, its exact-match performance is comparatively weak, with EMR values of 0.1527 at two qubits and 0.0519 at four qubits, showing that the re-uploading design requires sufficient Hilbert-space capacity to be effective.

For the BanglaBERT encoder, the classical baseline attains only a macro F1-score of 0.2101 and an EMR of 0.0000, and all three quantum circuits provide substantial gains.

SEL raises the macro F1-score to 0.4791 and the EMR to 0.3178 at four qubits, and to 0.3825 and 0.0264 at two qubits; at eight qubits the EMR falls back to 0.0387. The improvement is concentrated at the four-qubit level, where the single SEL best exploits the added capacity of the quantum subsystem.

VQC 1 improves steadily with the qubit size, reaching a macro F1-score of 0.4715 and an EMR of 0.3895 at eight qubits, with the partial-match F1-score rising from 0.5320 at two qubits to 0.5935 at eight qubits. Its EMR of 0.3891 at four qubits already exceeds the best value achieved by SEL, and it consistently outperforms the SEL circuit at every register size.

VQC 2 delivers the largest gains over the classical baseline: at four qubits it reaches a macro F1-score of 0.4850 and an EMR of 0.4071, and at eight qubits it attains the highest EMR of 0.4630 with a macro F1-score of 0.4798 and a partial-match F1-score of 0.6085. The trend in Fig. 6 shows that the re-uploading scheme exploits increasing qubit sizes most effectively, whereas the simpler encoding and entanglement strategies plateau or degrade.

Across all configurations in Table 3, the quantum branch consistently improves the Exact Match Ratio over the corresponding classical baseline whenever the encoding and entanglement structure are well matched to the encoder, with the largest joint-subset gains observed at the eight-qubit register. The full implications of these trends are examined in the Discussion.

5.4 Statistical Significance Analysis

To ascertain whether the observed performance differences between models are statistically meaningful, we conduct pairwise two-proportion z -tests on the aggregate Exact Match Ratio (EMR) over the $N = 2,272$ test samples. Table 4

Table 3: Performance comparison of IndicBERTv2 and BanglaBERT under different qubit and VQC configurations.

Framework	Qubit	VQC	Macro			Partial Match			Exact Match Ratio
			Precision	Recall	F1-score	Precision	Recall	F1-score	
IndicBERTv2	0	N/A (Classical)	0.4917	0.5900	0.5333	0.6136	0.6916	0.6268	0.4252
	2	SEL	0.2861	0.6220	0.3741	0.3564	0.7606	0.4739	0.0238
		VQC 1	0.4493	0.6406	0.4869	0.6039	0.6820	0.6164	0.4230
		VQC 2	0.3758	0.5720	0.4443	0.4601	0.7225	0.5386	0.1527
	4	SEL	0.4032	0.5796	0.4572	0.4825	0.6704	0.5387	0.2148
		VQC 1	0.4616	0.5496	0.4922	0.5996	0.6845	0.6141	0.4155
		VQC 2	0.3717	0.6115	0.4523	0.4062	0.7476	0.5156	0.0519
	8	SEL	0.3528	0.6212	0.4434	0.4082	0.7559	0.5189	0.0497
		VQC 1	0.4642	0.5521	0.4982	0.6030	0.6814	0.6144	0.4265
		VQC 2	0.5095	0.5301	0.5165	0.6266	0.6685	0.6255	0.4577
BanglaBERT	0	N/A (Classical)	0.1358	0.5000	0.2101	0.2716	0.7144	0.3870	0.0000
	2	SEL	0.2916	0.6392	0.3825	0.3637	0.7695	0.4822	0.0264
		VQC 1	0.3946	0.6033	0.4486	0.4521	0.7111	0.5320	0.1505
		VQC 2	0.1760	0.6667	0.2749	0.2640	0.9198	0.4039	0.0000
	4	SEL	0.4369	0.5713	0.4791	0.5407	0.6902	0.5796	0.3178
		VQC 1	0.4419	0.5136	0.4653	0.5710	0.6591	0.5858	0.3891
		VQC 2	0.4622	0.5174	0.4850	0.5954	0.6652	0.6062	0.4071
	8	SEL	0.3521	0.5994	0.4347	0.3974	0.7475	0.5093	0.0387
		VQC 1	0.4501	0.5001	0.4715	0.5809	0.6599	0.5935	0.3895
		VQC 2	0.4915	0.4780	0.4798	0.6175	0.6371	0.6085	0.4630

compares the classical baselines against each quantum configuration, and Table 5 presents register-scaling comparisons ($2 \rightarrow 4 \rightarrow 8$ qubits) within each circuit family.

Key findings from the significance analysis:

- **Statistical validity of quantum advantage under IndicBERTv2:** While Custom 1 is statistically indistinguishable from the classical baseline across all register sizes ($p > 0.45$), **Custom 2 at 8 qubits** achieves a statistically significant improvement over the classical baseline ($z = -2.21, p = 0.0274$). This confirms that the observed gain to an EMR of 0.4577 represents a genuine quantum enhancement rather than stochastic variance. Conversely, SEL is significantly worse than classical at every qubit width ($p < 0.0001$).
- **Compensatory effect under BanglaBERT:** As classical BanglaBERT collapses to zero EMR, all quantum setups with non-zero EMR achieve a statistically significant improvement ($p < 0.0001$). This validates the hypothesis that VQCs can act as compensatory nonlinear feature mappers when classical contextual embeddings underperform.
- **Register scaling patterns:** As shown in Table 5, the non-monotonic degradation of SEL from 4 to 8 qubits is statistically significant ($p < 0.0001$ on both encoders), confirming the under-entanglement bottleneck of a single SEL layer. Custom 1 demonstrates no significant change when expanding from 4 to 8 qubits ($p = 0.4527$ on IndicBERTv2, $p = 0.9779$ on BanglaBERT). Only Custom 2 exhibits a statistically significant positive scaling trajectory from 4 to 8 qubits ($p < 0.0001$ for IndicBERTv2, $p = 0.0001$ for BanglaBERT).

Table 4: Pairwise statistical significance comparison of classical baseline vs. each VQC configuration on Exact Match Ratio ($N = 2,272$, $\alpha = 0.05$).

Framework	Qubit	VQC Design	Classical vs. VQC EMR	z -statistic	p -value	Statistically Sig.?
IndicBERTv2	2	SEL	0.4252 vs. 0.0238	32.42	< 0.0001	Yes (Worse)
		Custom 1	0.4252 vs. 0.4230	0.15	0.8807	No
		Custom 2	0.4252 vs. 0.1527	20.26	< 0.0001	Yes (Worse)
	4	SEL	0.4252 vs. 0.2148	15.20	< 0.0001	Yes (Worse)
		Custom 1	0.4252 vs. 0.4155	0.66	0.5078	No
		Custom 2	0.4252 vs. 0.0519	29.52	< 0.0001	Yes (Worse)
	8	SEL	0.4252 vs. 0.0497	29.74	< 0.0001	Yes (Worse)
		Custom 1	0.4252 vs. 0.4265	-0.09	0.9294	No
		Custom 2	0.4252 vs. 0.4577	-2.21	0.0274	Yes (Better)
BanglaBERT	2	SEL	0.0000 vs. 0.0264	-7.85	< 0.0001	Yes (Better)
		Custom 1	0.0000 vs. 0.1505	-20.06	< 0.0001	Yes (Better)
		Custom 2	0.0000 vs. 0.0000	—	— [†]	N/A
	4	SEL	0.0000 vs. 0.3178	-32.53	< 0.0001	Yes (Better)
		Custom 1	0.0000 vs. 0.3891	-38.04	< 0.0001	Yes (Better)
		Custom 2	0.0000 vs. 0.4071	-39.50	< 0.0001	Yes (Better)
	8	SEL	0.0000 vs. 0.0387	-9.56	< 0.0001	Yes (Better)
		Custom 1	0.0000 vs. 0.3895	-38.07	< 0.0001	Yes (Better)
		Custom 2	0.0000 vs. 0.4630	-44.26	< 0.0001	Yes (Better)

[†] Both models attain an EMR of 0.0000; no difference exists to test.

Table 5: Pairwise statistical significance of adjacent register-size scaling ($2 \rightarrow 4 \rightarrow 8$ qubits) within each VQC architecture on Exact Match Ratio.

Framework	VQC Architecture	Comparison	EMR Transition	z -statistic	p -value	Significant?
IndicBERTv2	SEL	2q vs. 4q	0.0238 vs. 0.2148	-19.86	< 0.0001	Yes
		4q vs. 8q	0.2148 vs. 0.0497	16.43	< 0.0001	Yes
	Custom 1	2q vs. 4q	0.4230 vs. 0.4155	0.51	0.6084	No
		4q vs. 8q	0.4155 vs. 0.4265	-0.75	0.4527	No
	Custom 2	2q vs. 4q	0.1527 vs. 0.0519	11.21	< 0.0001	Yes
		4q vs. 8q	0.0519 vs. 0.4577	-31.39	< 0.0001	Yes
BanglaBERT	SEL	2q vs. 4q	0.0264 vs. 0.3178	-26.02	< 0.0001	Yes
		4q vs. 8q	0.3178 vs. 0.0387	24.58	< 0.0001	Yes
	Custom 1	2q vs. 4q	0.1505 vs. 0.3891	-18.12	< 0.0001	Yes
		4q vs. 8q	0.3891 vs. 0.3895	-0.03	0.9779	No
	Custom 2	2q vs. 4q	0.0000 vs. 0.4071	-39.50	< 0.0001	Yes
		4q vs. 8q	0.4071 vs. 0.4630	-3.80	0.0001	Yes

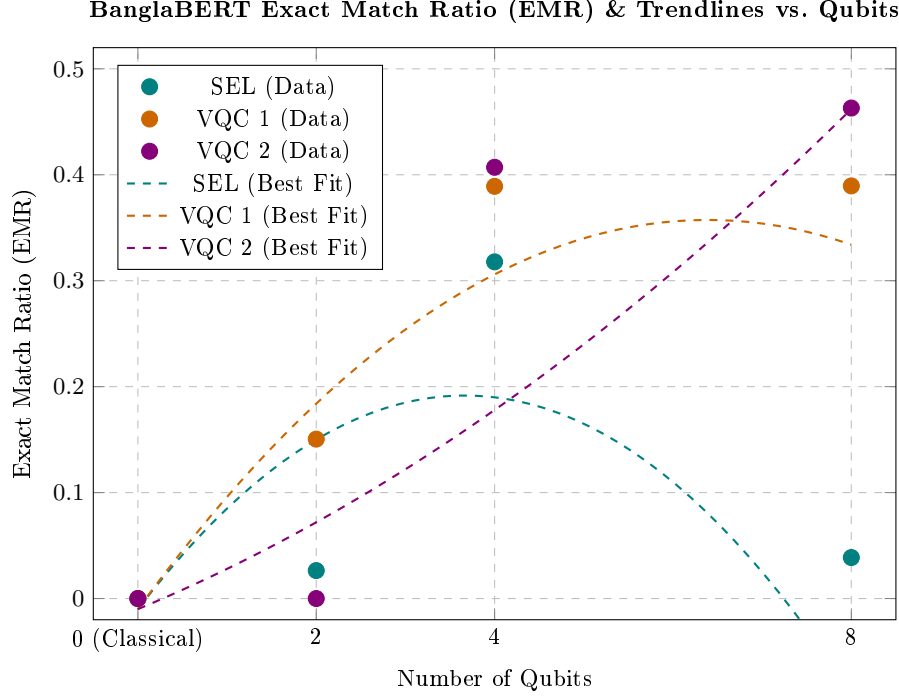


Fig. 6: BanglaBERT EMR across qubit counts for SEL, VQC 1, and VQC 2, together with their fitted trend lines.

5.5 Single-Label Subset Evaluation and Class Collapse

To examine whether VQC architectures preserve distinct decision boundaries or suffer from majority-class collapse, we evaluate models on the single-label test subset ($N_{\text{single}} = 940$, label cardinality $\sum_k y_{ik} = 1$). This subset comprises six emotion categories: *Happy* ($N = 300, 31.9\%$), *Sad* ($N = 200, 21.3\%$), *Angry* ($N = 200, 21.3\%$), *Disgust* ($N = 100, 10.6\%$), *Surprise* ($N = 80, 8.5\%$), and *Fear* ($N = 60, 6.4\%$). Table 6 summarizes the single-label EMR benchmarks.

Architectural failure modes and boundary separability:

- **SEL Architecture:** While SEL records moderate single-label EMRs (0.4713 on IndicBERTv2, 0.4500 on BanglaBERT), fine-grained error analysis reveals severe majority-class bias. SEL yields non-zero recall almost exclusively on high-support classes (*Happy* and *Angry*), while completely failing on minority classes (*Disgust*, *Surprise*, and *Fear* all exhibit 0.0000 recall).
- **Custom 1 Architecture:** In the absence of data re-uploading, Custom 1 undergoes catastrophic class collapse under the weaker BanglaBERT encoder. The model predicts the majority *Happy* label for every single sample (100% recall on *Happy*, 0% recall on all other 5 classes). Consequently, its single-label

Table 6: Single-label Exact Match Ratio (EMR) across evaluated VQC architectures on the mutually exclusive test partition ($N_{\text{single}} = 940$).

Encoder	VQC Architecture	Qubits	Exact Match Ratio (EMR)
IndicBERTv2	SEL reference	4	0.4713
IndicBERTv2	Custom 1 (Cross-coupling)	4	0.4521
IndicBERTv2	Custom 2 (Re-uploading)	8	0.5479
BanglaBERT	SEL reference	4	0.4500
BanglaBERT	Custom 1 (Cross-coupling)	4	0.3191
BanglaBERT	Custom 2 (Re-uploading)	8	0.5330

EMR exactly equals the proportion of *Happy* instances in the single-label test split ($300/940 = 0.3191$).

- **Custom 2 Architecture:** By combining learnable per-qubit phase (ϕ_q) and scale (s_q) input rotations with data re-uploading ($U(x) \rightarrow V(\theta_1) \rightarrow U(x) \rightarrow V(\theta_2)$) and expanded 16-channel Pauli-Z/X expectation readouts, Custom 2 builds highly non-linear decision boundaries. It achieves peak single-label EMR scores of **0.5479** (IndicBERTv2) and **0.5330** (BanglaBERT), successfully predicting all minority classes and preventing majority-class collapse across both language models.

5.6 Empirical Runtime and Latency Analysis

Table 7 provides an empirical latency breakdown recorded during training runs on the PennyLane statevector simulator (`default.qubit`, CPU execution).

Table 7: Measured variational quantum circuit runtime and latency profiling across representative configurations (batch size $B = 8$, CPU execution).

VQC Architecture	Qubits	Encoder	Total VQC Time	Forward Calls	Per Batch (ms)	Per Sample (ms)
SEL reference	4	IndicBERTv2	67.15 min (4029.1 s)	59,141	68.13	8.52
SEL reference	4	BanglaBERT	82.50 min (4950.2 s)	59,141	83.70	10.46
Custom 1 (Cross-coupling)	4	IndicBERTv2	89.58 min (5374.5 s)	64,259	83.64	10.45
Custom 1 (Cross-coupling)	4	BanglaBERT	87.31 min (5238.4 s)	59,141	88.58	11.07
Custom 2 (Run A)	8	IndicBERTv2	227.33 min (13639.8 s)	48,905	278.90	34.86
Custom 2 (Run B)	8	IndicBERTv2	195.49 min (11729.2 s)	41,228	284.50	35.56

Latency observations and scalability insights:

- **Empirical validation of 2^{n_q} scaling:** At 4 qubits ($2^4 = 16$ statevector dimension), per-sample latency ranges between 8.52 ms and 11.07 ms across SEL and Custom 1. Doubling the register size to 8 qubits ($2^8 = 256$ statevector dimension) in Custom 2 increases per-sample execution time to ≈ 35 ms ($\approx 3.3\times$ increase). The close agreement between Run A (34.86 ms) and Run B (35.56 ms) confirms high runtime reproducibility.

- **Circuit depth overhead:** Custom 1 incurs only modest runtime overhead compared to SEL (10.45 ms vs. 8.52 ms on IndicBERTv2), reflecting the addition of single-qubit rotations while retaining a 16-dimensional statevector.
- **Hybrid system bottleneck:** Although statevector simulation cost grows exponentially with n_q , the classical forward pass through the transformer encoder (IndicBERTv2, ≈ 278 M parameters) remains the dominant computational component of the end-to-end training epoch. Hence, 8-qubit simulation remains fully practical for near-term natural language classification pipelines.

6 Discussion

The results demonstrate that both the pretrained encoder and the quantum-circuit design strongly influence classification performance. IndicBERTv2 provides the strongest classical representation, whereas the quantum variants produce particularly large gains for the BanglaBERT baseline. As the qubit count increases, VQC 1 shows relatively consistent improvement, while VQC 2 achieves the highest EMR for both encoders, reaching 0.4577 with IndicBERTv2 and 0.4630 with BanglaBERT, and also records the largest single gain over the classical baseline (EMR 0.4630 versus 0.0000 for BanglaBERT). SEL, by contrast, peaks at the four-qubit configuration, attaining its best EMR of 0.3178 with BanglaBERT, and degrades at two and eight qubits. The trend in Fig. 6 further indicates that additional qubits are most beneficial when paired with an expressive encoding and entanglement strategy; increasing circuit size alone does not guarantee uniform improvement across all configurations.

Encoder-circuit interaction. A striking asymmetry emerges between the two pretrained encoders. Under IndicBERTv2, the classical baseline already achieves a strong macro F1-score of 0.5333 and an EMR of 0.4252, leaving relatively little room for improvement; the best quantum configuration (VQC 2, eight qubits) raises the EMR by only 0.0325 absolute points. Under BanglaBERT, however, the classical baseline collapses to an EMR of 0.0000 and a macro F1-score of 0.2101, and the same VQC 2 configuration recovers an EMR of 0.4630, which is an absolute gain of 0.4630 points. This pattern suggests that the variational quantum branch acts as a compensatory mechanism: when the classical feature extractor already produces well-separated class representations, the additional nonlinearity of the quantum circuit adds only marginal benefit, but when the classical embedding is weak, the quantum subsystem can learn discriminative rotations and entangled correlations that the linear projection alone cannot capture. Crucially, this interaction is architecture-dependent: SEL provides only partial compensation (EMR 0.3178 at four qubits), whereas the richer entangling and re-uploading strategies of VQC 1 and VQC 2 exploit the compensatory role more fully.

Role of data re-uploading. The data re-uploading strategy employed by VQC 2, in which the classical feature encoding $U(x)$ is inserted before each of the two

variational blocks, proves to be the most effective design choice in this study. Pérez-Salinas et al. [13] showed theoretically that re-uploading enables a single qubit to approximate any function on the input sphere; in the multi-qubit setting, this translates to a richer landscape of decision boundaries. The empirical evidence supports this: Custom 2 achieves the highest EMR at the eight-qubit register for both encoders (0.4577 and 0.4630), and its trend line in Fig. 6 is the only one with a consistently positive slope, indicating that additional qubits continue to improve performance rather than saturating. By contrast, VQC 1, which uses a single encoding layer followed by two variational blocks, plateaus in EMR at four and eight qubits for BanglaBERT (0.3891 versus 0.3895). The re-uploading mechanism thus appears to provide a form of “quantum feature amplification” that allows the circuit to re-examine the input at each layer and build progressively more complex correlations.

Implications for NISQ deployment. All experiments in this study were conducted on the PennyLane `default.qubit` simulator, which provides exact statevector simulation without shot noise or hardware errors. On current noisy intermediate-scale quantum (NISQ) devices, gate fidelities and qubit coherence times would impose additional constraints. The one layer for SEL and two layers for VQC 1 and VQC 2 are well suited to NISQ hardware because their circuit depth remains modest. VQC 2’s use of controlled-Z gates rather than CNOT gates may offer an additional advantage on superconducting architectures where CZ is a native gate, reducing the need for gate decomposition and thereby lowering the effective error rate. The four-qubit register, which achieves competitive EMR values across all three circuits, represents a practical entry point for near-term hardware experiments, as current quantum processors routinely support four fully connected qubits with gate fidelities above 99%.

Class imbalance and quantum capacity. The EmoNoBa dataset exhibits extreme label imbalance (*Joy* 45%, *Fear* 1.5%), which interacts with the quantum circuit in two ways. First, frequency-based `pos_weight` values in the loss up-weight rare-emotion gradients, steering quantum parameters toward minority-class decision boundaries. VQC 2 exploits these amplified gradients most effectively, whereas the classical-only BanglaBERT model fails to predict Fear at all. Second, per-emotion threshold calibration on the validation set is essential for converting the quantum model’s improved recall into exact-match accuracy; without it, sigmoid outputs favour high-frequency labels and suppress rare ones, negating the quantum branch’s benefit.

Statistical validation of quantum enhancement. Hypothesis testing on aggregate EMR confirms that the performance differences are not artifacts of sampling variance. Under IndicBERTv2, Custom 2 at 8 qubits is the sole quantum architecture that statistically outperforms the strong classical baseline ($p = 0.0274$), demonstrating that a sufficiently expressive ansatz operating in an expanded 256-dimensional Hilbert space provides genuine representational value beyond

classical linear transformations. In contrast, Custom 1 remains statistically indistinguishable from the classical baseline ($p > 0.45$), confirming that simple fixed-pattern entanglement without data re-uploading fails to capture higher-order feature interactions. Under BanglaBERT, where the classical baseline collapses, all non-zero quantum models achieve high statistical significance ($p < 0.0001$), solidifying the compensatory role of quantum embeddings for under-trained or low-capacity language backbones.

Single-label boundary resolution versus class collapse. The single-label benchmark ($N_{\text{single}} = 940$) reveals a stark contrast in geometric separability across VQC designs. Custom 1 suffers total majority-class collapse under BanglaBERT (predicting *Happy* for all samples, producing an EMR of exactly 0.3191), while SEL exhibits severe majority bias with complete zero-recall failure on minority classes. By interleaving data encoding with parameterized unitaries and using learnable phase-scale rotations, Custom 2 avoids majority collapse and establishes well-separated hyperplanes, achieving peak single-label EMRs of 0.5479 and 0.5330. This indicates that data re-uploading is vital not only for multi-label subset prediction but also for resolving mutually exclusive emotional boundaries under extreme class imbalance.

Computational overhead versus representation trade-off. Empirical latency profiling validates the theoretical $T_{\text{batch}} = \mathcal{O}(Bn_q2^{n_q})$ scaling of statevector simulation, showing a jump from ≈ 10 ms per sample at 4 qubits to ≈ 35 ms at 8 qubits. Although this $3.3\times$ latency increase is non-trivial, it remains computationally subordinate to the classical 278M-parameter transformer forward pass during end-to-end training. Given the statistically significant exact-match improvements achieved by 8-qubit Custom 2, this overhead represents an advantageous computational trade-off for hybrid quantum NLP workflows.

7 Conclusion

This paper investigated hybrid quantum-classical frameworks for multi-label emotion classification of Bengali text. Three VQCs were designed and evaluated under an identical outer framework: the SEL VQC, which combines a simple R_X angle encoding with a single layer of PennyLane’s **StronglyEntanglingLayers** and Pauli-Z readout; the VQC 1 cross-coupling VQC, which combines a simple R_Y angle encoding with two entangling layers using distinct nearest-neighbour and cross-coupling CNOT topologies; and the VQC 2 re-uploading VQC, which combines R_X/R_Y encoding with learnable per-qubit phase and scale parameters, a rotating ansatz, and input re-injection between variational layers.

Three main conclusions follow from the experimental study. First, the learned representation space depends strongly on the selected classical encoder and quantum embedding. The IndicBERTv2 classical baseline achieved a macro F1-score of 0.5333, compared with 0.2101 for BanglaBERT. The strongest quantum configurations reached macro F1-scores of 0.5165 for IndicBERTv2 and 0.4850 for

BanglaBERT, while the highest exact-match ratios were 0.4577 and 0.4630, respectively, all achieved by the re-uploading VQC 2 design. The SEL VQC, by contrast, improved only the BanglaBERT baseline, raising its exact-match ratio to 0.3178 at four qubits, while never surpassing the classical model under IndicBERTv2. These results show that the quality of the pretrained representation remains important, although the quantum configurations substantially improve the BanglaBERT baseline.

Second, scaling behaviour is architecture- and encoder-dependent. The VQC 1 macro F1-score improves consistently as the qubit count grows from two to eight qubits for both IndicBERTv2 (0.4869 to 0.4982) and BanglaBERT (0.4486 to 0.4715). VQC 2 exhibits a larger but less uniform gain, with its strongest performance appearing at eight qubits for IndicBERTv2 and four qubits for BanglaBERT; at the eight-qubit configuration, its exact-match ratio exceeds that of VQC 1 for both encoders (0.4577 versus 0.4265 and 0.4630 versus 0.3895). SEL shows the least favourable scaling: its benefit is concentrated at four qubits and degrades sharply at two and eight qubits, indicating that a single SEL does not exploit larger qubit counts. This suggests that additional Hilbert-space capacity is useful only when the encoding, entanglement pattern, and classical representation are jointly compatible, and that the re-uploading scheme exploits this capacity most effectively.

Third, simulator-backend selection does not alter the reported analytic scores. The PennyLane `default.qubit` simulator is used for all reported results, and switching between mathematically equivalent simulator backends reduces wall-clock training time from several hours to tens of minutes without changing the model outputs. The experiments also indicate that a four-qubit circuit trained with adjoint differentiation is preferable to backpropagation on PennyLane’s accelerated devices. AdamW optimization, plateau-based learning-rate reduction, and early stopping provide a stable training protocol for the hybrid model on the strongly imbalanced EmoNoBa corpus.

8 Limitations and Future Work

The principal limitation is that the quantum branch introduces relatively few trainable parameters, so its benefit depends heavily on the quality of the compressed classical features. For very small qubit configurations, the quantum contribution can be dominated by the projection and post-processing layers. In addition, all experiments were conducted on quantum simulators rather than on physical hardware. Future work should extend the comparison to larger registers with controlled re-uploading depth, combine cross-coupling entanglement with data re-uploading in a single ansatz, incorporate multi-observable and correlated readouts, and evaluate the strongest architectures under finite-shot and noisy simulation before deployment on quantum hardware.

A further limitation concerns the evaluation protocol itself. The EMR is an intentionally strict criterion that assigns a score of zero to any sample for which the predicted label set differs from the gold label set in even a single emotion,

and the reported partial-match scores are macro-averaged over instances without instance-level confidence weighting. Both measures therefore depend on the choice of the decision threshold, and the per-label thresholds used here were tuned on the validation partition, which may not transfer perfectly to other data distributions. The dataset also spans a single language and a limited set of social-media domains, so the conclusions may not generalise to other low-resource languages, longer texts, or formal registers without retraining.

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